# Akoizo Original Practice - 2027 Rotation Form 4 - Answer key

### Division C | Chemistry Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) At constant temperature, a gas at 1.8 atm occupies 8 L. Compress it to 4 L. Find the final pressure.**

answer: 3.6 atm (2 points).

Numerical target 3.6; absolute tolerance 0.0072. Accepted units: , atm.

**2. (2 points) At constant pressure, a gas occupies 8 L at 380 K. Find its volume at 760 K.**

answer: 16 L (2 points).

Numerical target 16; absolute tolerance 0.032. Accepted units: , L.

**3. (2 points) A rigid vessel contains a gas at 1.8 atm and 380 K. Find its pressure at 190 K with no gas lost.**

answer: 0.9 atm (2 points).

Numerical target 0.9; absolute tolerance 0.0018. Accepted units: , atm.

**4. (2 points) Use PV=nRT with R=0.08206 L atm/(mol K). P=1.8 atm, V=8 L and T=380 K. Find the amount of gas.**

answer: 0.4617930397526842 mol (2 points).

Numerical target 0.46179304; absolute tolerance 0.00092358608. Accepted units: , mol.

**5. (2 points) A dry gas mixture has partial pressures 1.8 atm and 0.9 atm. Find total pressure.**

answer: 2.7 atm (2 points).

Numerical target 2.7; absolute tolerance 0.0054. Accepted units: , atm.

**6. (2 points) A mixture has 6 mol oxygen and 18 mol nitrogen at total pressure 8 atm. Find oxygen partial pressure.**

answer: 2 atm (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , atm.

**7. (2 points) Gas collected over water has total pressure 104 kPa. Water-vapor pressure is supplied as 7 kPa. Find dry-gas pressure.**

answer: 97 kPa (2 points).

Numerical target 97; absolute tolerance 0.194. Accepted units: , kPa.

**8. (2 points) Two ideal gases at the same temperature have molar masses 4 and 16 g/mol. What is the diffusion-rate ratio light/heavy? Use Graham's law.**

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , .

**9. (2 points) For 2 H2 + O2 -> 2 H2O, 6 mol O2 reacts completely with excess H2. How many moles of H2 are consumed?**

answer: 12 mol (2 points).

Numerical target 12; absolute tolerance 0.024. Accepted units: , mol.

**10. (2 points) Explain why gas-law temperature ratios must use kelvin rather than degrees Celsius.**

answer: Kelvin is proportional to absolute thermodynamic temperature. Celsius has an arbitrary offset; its ratios do not represent molecular kinetic-energy ratios. (2 points).

**11. (2 points) [State/National] Determine the order with respect to A from the supplied initial-rate data.**

Synthetic initial-rate data (M/s): [A]=0.18 M, [B]=0.10 M gives rate 0.01944; [A]=0.36, [B]=0.10 gives 0.07776; [A]=0.18, [B]=0.20 gives 0.03888. Assume rate=k[A]^m[B]^n.

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , .

**12. (2 points) [State/National] Determine the order with respect to B from the supplied initial-rate data.**

Synthetic initial-rate data (M/s): [A]=0.18 M, [B]=0.10 M gives rate 0.01944; [A]=0.36, [B]=0.10 gives 0.07776; [A]=0.18, [B]=0.20 gives 0.03888. Assume rate=k[A]^m[B]^n.

answer: 1 (2 points).

Numerical target 1; absolute tolerance 0.002. Accepted units: , .

**13. (2 points) [State/National] Calculate k from the initial-rate data. Give its value in M^-2 s^-1.**

Synthetic initial-rate data (M/s): [A]=0.18 M, [B]=0.10 M gives rate 0.01944; [A]=0.36, [B]=0.10 gives 0.07776; [A]=0.18, [B]=0.20 gives 0.03888. Assume rate=k[A]^m[B]^n.

answer: 6 M^-2 s^-1 (2 points).

Numerical target 6; absolute tolerance 0.012. Accepted units: , M^-2 s^-1.

**14. (2 points) [State/National] Predict the rate at [A]=0.54 M and [B]=0.10 M.**

Synthetic initial-rate data (M/s): [A]=0.18 M, [B]=0.10 M gives rate 0.01944; [A]=0.36, [B]=0.10 gives 0.07776; [A]=0.18, [B]=0.20 gives 0.03888. Assume rate=k[A]^m[B]^n.

answer: 0.17496 M/s (2 points).

Numerical target 0.17496; absolute tolerance 0.00034992. Accepted units: , M/s.

**15. (2 points) [State/National] For a first-order reaction, k=0.0495 s^-1. Use ln(2)=0.693 to calculate the half-life.**

answer: 14 s (2 points).

Numerical target 14; absolute tolerance 0.028. Accepted units: , s.

**16. (2 points) [State/National] A first-order reactant starts at 1.12 M. Find concentration after three half-lives.**

answer: 0.14 M (2 points).

Numerical target 0.14; absolute tolerance 0.00028. Accepted units: , M.

**17. (2 points) [State/National] A zero-order reaction starts at 1.4 M and consumes reactant at 0.02 M/s. Find concentration after 10 s.**

answer: 1.2000000000000002 M (2 points).

Numerical target 1.2; absolute tolerance 0.0024. Accepted units: , M.

**18. (2 points) A catalyst increases rate without changing reactant/product free energies. Explain the mechanism and whether the equilibrium constant changes.**

answer: It offers a lower-activation-energy pathway and speeds approach to equilibrium. At fixed temperature it does not change the equilibrium constant. (2 points).

**19. (2 points) Equal masses of the same solid react with excess acid under matched conditions. Why can smaller particles react faster?**

answer: Smaller particles expose more surface area, increasing contact and collision opportunities; mass alone does not set exposed area. (2 points).

**20. (2 points) [State/National] At 300 K a rate constant is k1; at 330 K it is 2k1. Using R=8.314 J/(mol K), ln(2)=0.693 and ln(k2/k1)=Ea/R(1/T1-1/T2), find Ea.**

answer: 19013.286599999985 J/mol (2 points).

Numerical target 19013.287; absolute tolerance 38.026573. Accepted units: , J/mol.